

Reinforcement Learning

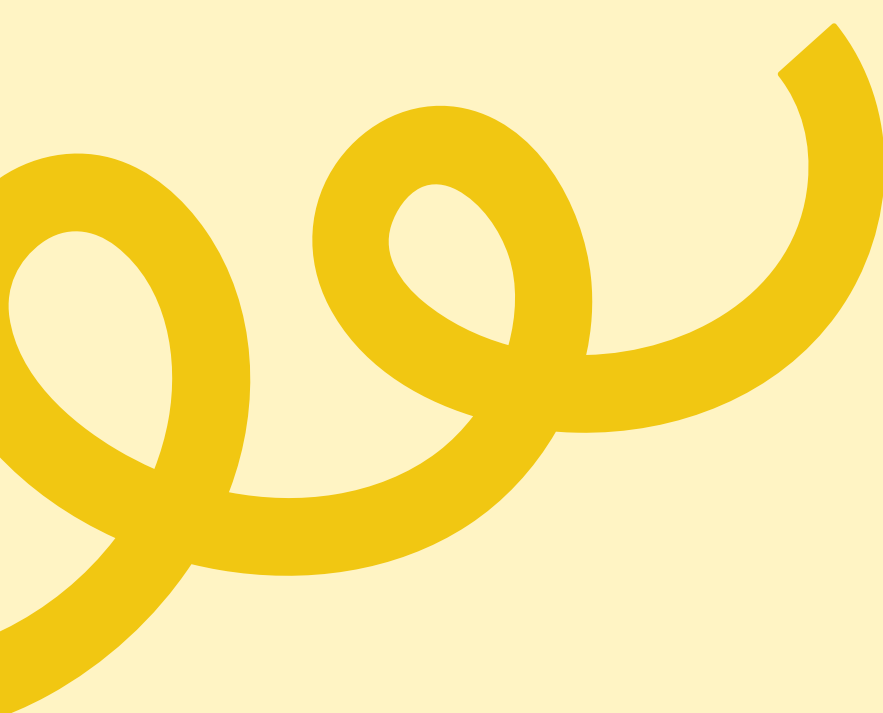
Reward & Penalty Game



What is Reinforcement Learning?

Reinforcement Learning is a way to learn by trying things and getting rewards or penalties. When you try something new, you find out if it works or not! Learning happens by doing and seeing what happens next. Just like when you learn to ride a bike - you keep trying until you get it right!





How Do We Learn?

When we do something right, we get a reward (like a smile or prize). When we do something wrong, we get a penalty (like a timeout or no points). This helps us remember what to do next time!





Real-Life Examples Kids Know



Learning to ride a bike

Reward: Balance and riding smoothly!

Penalty: Falling down (ouch!)



Training a pet

Reward: Yummy treats for good tricks!

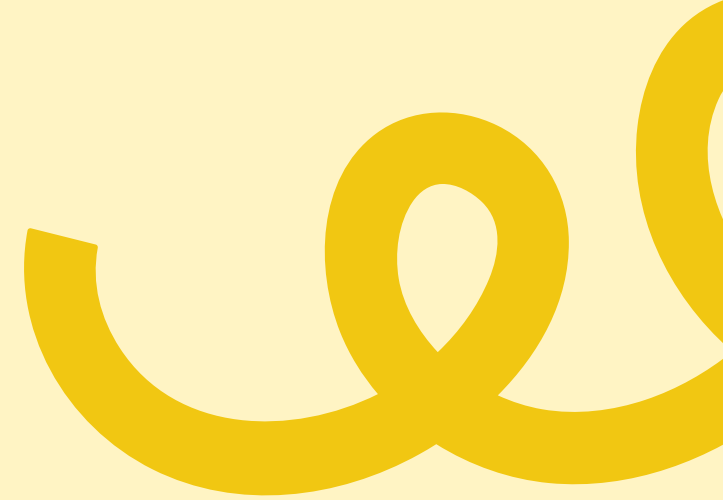
Penalty: No treat when they don't listen



Video games

Reward: Points and level up!

Penalty: Lose a life or game over



The Reward & Penalty Game



Imagine a game where you get points for good moves and lose points for mistakes. The goal is to learn how to get more points by playing smart! Every action you take teaches you something new about winning.

How Do Computers Learn Like Us?

Computers learn just like we do! They try actions and get rewards or penalties.

- Humans: Learn by practice and feedback
- Computers: Learn by trying many times very fast
- Humans: Remember what worked before
- Computers: Store information in memory
- Humans: Get better with experience
- Computers: Improve by playing games over and over



Meet the Agent and Environment



Agent: The player or computer trying to learn something new.

Environment: The world or game where the agent acts and explores.

Think of yourself as the agent and your playground as the environment!



Actions, Rewards & Penalties

Actions: What the agent decides to do, like moving left or jumping!

Rewards: Points or prizes for good actions, like getting a gold star!

Penalties: Points lost or small punishments for wrong actions, like losing a turn.

These three things help us learn what works best!



Interactive Game Example



Imagine playing a ball-catching game! Your actions are moving left or right. Catch the ball = Reward (points!). Miss the ball = Penalty (lose points). Think: What would you do differently next time to catch more balls?

Fun Activity: Reward or Penalty?

Let's play a game! Your teacher will read an action, and you decide if it deserves a reward or penalty!

- "You finish your homework early" → Reward or Penalty?
 - "You shout loudly in class" → Reward or Penalty?
 - "You help a friend who is stuck" → Reward or Penalty?
 - "You forget to say thank you" → Reward or Penalty?
- Think carefully and raise your hand to answer!



Positive Reinforcement

Positive reinforcement means giving a reward to encourage good behavior. When you do something good, you get something nice in return!
Example: Getting a shiny sticker for cleaning your room makes you want to keep your room tidy again!
Rewards help us remember to repeat good actions.
It makes learning fun and exciting!



Negative Reinforcement

Negative reinforcement means removing something unpleasant when good behavior happens. It's not a punishment! Instead, it takes away something annoying or uncomfortable as a reward.

Example: When you finish your homework, your parents stop reminding you about it. The nagging stops because you did something good!

Another example: When you clean your room, you don't have to hear "Please clean your room" anymore. That feels like a relief!



Trial and Error Learning

Learning happens by trying different actions until you find the best one. Sometimes we fall, sometimes we miss, but that's okay! Every mistake teaches us something new. It's okay to make mistakes because that helps us learn and grow. The more we try, the better we become!



Making Smart Choices



Think before you act - pause and consider your options!

Learn from what worked before - remember your past successes.

Ask yourself: What reward do I want to get?

If one way doesn't work, try a different way.

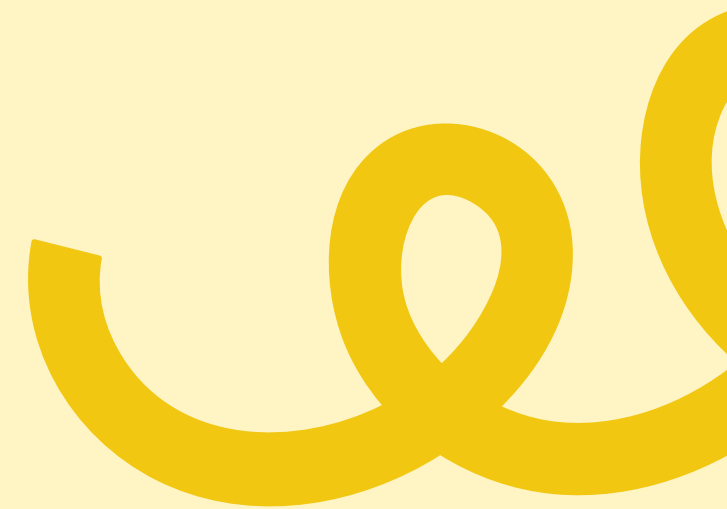
Practice makes your choices even smarter over time!



Practice Makes Perfect



Keep trying the reward & penalty game to get better each time! Play fun games that help you learn new skills. Practice with your friends or family members. The more you practice, the smarter your choices become. Remember, every try teaches you something new!





Summary & Key Takeaways



- Reinforcement learning helps us learn with rewards and penalties.
 - Computers learn like us by trying actions and getting feedback.
 - Practice and smart choices help us get better.
 - Keep having fun while learning!
- Thank you! Any questions?

